INDIRA GANDHI CENTRE FOR ATOMIC RESEARCH

Department of Atomic Energy, Government of India

PROJECT REPORT

on

Development of Physics-Informed Neural Network Model for Core Neutronics in the Context of PFBR Operator Training Simulator

Submitted by

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I am deeply grateful to my project supervisor and mentor, **Shri P. Raghava Reddy**, for his continuous guidance, technical insights, constructive feedback, and encouragement throughout the course of this work. His expertise in reactor physics, computational modelling, and scientific computing greatly contributed to my understanding of the problem and to the successful development of the proposed machine-learning approach.

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I express my sincere gratitude to my faculty mentor, **Dr. Hoshiyar Singh Kanyal**, along with the faculty members and the Department of Computer Science and Engineering at ABES Institute of Technology, affiliated with Dr. A.P.J. Abdul Kalam Technical University (AKTU), for their academic guidance and support throughout my undergraduate studies.

Finally, I am grateful to my family and friends for their constant encouragement, support, and motivation.This project has been a valuable learning experience, enabling me to strengthen my knowledge of deep learning, recurrent neural networks, physics-informed machine learning, reactor kinetics, and scientific computing, while also providing an opportunity to understand the practical application of computational techniques in a research-oriented environment.

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INTRODUCTION:

The Indira Gandhi Centre for Atomic Research (IGCAR) is a premier research and development centre under the Department of Atomic Energy (DAE), Government of India, located at Kalpakkam, Tamil Nadu. Established in 1971, the centre has a major focus on multidisciplinary research and advanced engineering for the development of sodium-cooled Fast Breeder Reactor (FBR) technology in India.

IGCAR has made significant contributions to India's fast reactor programme through the development and operation of major nuclear research facilities. The Fast Breeder Test Reactor (FBTR) serves as an important facility for research and testing of fast-reactor fuels, materials, and components. IGCAR's work with FBTR has provided valuable experience in fast-reactor operation and technology development.

Another major achievement is the development of the Prototype Fast Breeder Reactor (PFBR), a 500 MWe sodium-cooled fast breeder reactor designed as an important step in India's advanced nuclear power programme. The PFBR represents the application of decades of research and development in reactor physics, fuel technology, materials, sodium systems, safety, instrumentation, and reactor engineering at IGCAR.

IGCAR also operates the KAlpakkam MINI Reactor (KAMINI), a 30 kW thermal research reactor using U-233 fuel. KAMINI provides neutron irradiation and research facilities and is used for applications such as neutron activation analysis and neutron radiography.

In addition to reactor technology, IGCAR conducts research in areas including reactor physics, materials science, thermal hydraulics, instrumentation and control, computational modelling, and simulation. The centre's research facilities and simulators also provide opportunities for students and researchers to work on advanced scientific and engineering problems.

The present project is carried out in this research environment and focuses on the application of machine learning and physics-informed modelling to reactor kinetics associated with the PFBR Operator Training Simulator. The project explores the use of recurrent neural networks, particularly LSTM, to model the temporal behaviour of reactor power and delayed neutron precursor concentrations while incorporating the underlying physical constraints of reactor kinetics.

ABSTRACT:

Nuclear reactor operation requires accurate modelling of complex physical processes and reliable prediction of important plant parameters. Operator Training Simulators (OTS) provide a controlled environment in which reactor operators can study plant behaviour and practise operating procedures without interacting with the actual reactor. IGCAR has developed expertise in fast breeder reactor technology and in the development of full-scope operator training simulators for fast breeder reactors.

This internship focused on the development of a Physics-Informed Neural Network (PINN) model for core neutronics in the context of the Prototype Fast Breeder Reactor (PFBR) Operator Training Simulator. Machine learning (ML) offers the ability to learn nonlinear relationships from data and can potentially provide computationally efficient surrogate models for selected scientific and engineering calculations. However, purely data-driven models can become unreliable outside the distribution of their training data and may not explicitly respect known physical laws. PINNs address this limitation by incorporating governing equations and physical constraints into the learning objective.

The work involved studying machine learning fundamentals, neural networks, reactor kinetics and delayed-neutron precursor behaviour, formulating a physics-informed learning problem, preparing reactor-related time-series data, designing a neural network model, and combining data and physics information in the training objective. The working dataset considered time, reactivity and reactor-state quantities including power and six delayed-neutron precursor variables. The target formulation used time as the model input, reactivity as an input to the physics-based loss, and reactor-state quantities as predicted outputs.

The overall objective was to investigate whether a physics-informed neural model can reproduce the expected reactor dynamics with sufficiently small prediction error for potential use as a computational component within a simulator framework. The report presents the background, methodology, model formulation, implementation, evaluation strategy, limitations, and possible future extensions of the work.

TABLE OF CONTENTS:

1. PROJECT INTRODUCTION:

1.1 Background

Machine Learning (ML) is a branch of artificial intelligence in which computational models learn patterns or relationships from data and use the learned relationships for prediction, classification, optimisation, or decision support. Unlike a conventional program in which every relationship must be explicitly encoded, an ML model adjusts its internal parameters during training so that its output approaches the desired target.

Modern neural networks are particularly useful for problems involving nonlinear and high-dimensional relationships. In engineering, they can be trained using measurements, simulation outputs, historical operating data, or combinations of these sources. Their ability to approximate nonlinear functions makes them attractive for scientific computing, surrogate modelling, anomaly detection, forecasting, and optimisation.

1.2 Why Machine Learning is Relevant to Nuclear Reactors

Nuclear reactors are complex physical systems involving coupled neutronic, thermal-hydraulic, structural, chemical, instrumentation, and control processes. Traditional reactor analysis relies on well-established mathematical and numerical models. These models are essential because they are grounded in physical laws and are used for safety analysis, design, operation, and licensing. At the same time, some high-fidelity calculations can be computationally expensive, while plant operation generates large amounts of time-dependent sensor and process data.

Machine learning can complement established reactor modelling rather than replace it. Potential applications include prediction of reactor parameters, anomaly and fault detection, sensor validation, condition monitoring, predictive maintenance, surrogate modelling of computationally expensive calculations, design optimisation, and decision-support systems. Reviews of AI applications in nuclear reactors identify reactor monitoring, operation and maintenance, fault detection and diagnosis, optimisation, and prediction as important application areas.

1.3 Machine Learning in Operator Training Simulators

An Operator Training Simulator is designed to reproduce important plant behaviours in a controlled simulation environment so that operators can practise normal operation, transients, abnormal situations, and procedural responses. IGCAR's Instrumentation and Control Division identifies development of operator training simulators for fast breeder reactors among its areas of expertise, and IGCAR has reported development and commissioning of a full-scope replica PFBR operator training simulator.

IGCAR's published material on PFBR simulator development describes the use of numerical models to represent physical processes and highlights the need for simulation time to be compatible with real-time evolution. This creates an opportunity to investigate ML-based surrogate models for selected calculations, provided that accuracy, stability, physical consistency, validation, and safety requirements are satisfied.

1.4 PFBR Operator Training Simulator

A full-scope operator training simulator reproduces important plant systems and their dynamic behaviour to support operator training. IGCAR's technical material identifies major simulator subsystems including neutronics, primary sodium, secondary sodium, steam-water, and electrical systems. The simulator therefore provides an environment in which plant dynamics can be studied without exposing personnel or equipment to actual reactor operation.

1.5 Motivation for Physics-Informed Learning

A purely data-driven neural network can fit observed examples well but may behave unpredictably for inputs outside the training distribution. This is particularly important in nuclear engineering, where physical consistency and reliable behaviour are essential. Physics-Informed Neural Networks address this issue by including governing equations in the training objective. The PINN framework introduced by Raissi, Perdikaris, and Karniadakis combines data-driven learning with mathematical constraints and can be useful in settings where data are limited or incomplete.[1]

1.6 Role of Core Neutronics

Core neutronics describes the behaviour of neutrons within the reactor core and the resulting fission chain reaction. Reactor power is strongly linked to neutron population, while reactivity and delayed neutron precursors influence the time evolution of the reactor. A reduced-order kinetics model can therefore provide an important representation of reactor dynamic behaviour for simulator applications.

1.7 Project

The internship project was titled "Development of Physics-Informed Neural Network Model for Core Neutronics in the Context of PFBR Operator Training Simulator." The central idea was to investigate a neural-network model that learns reactor-state evolution

Objective	Description
Understand reactor context	Study PFBR, core neutronics, reactor kinetics, and the role of an operator training simulator.
Study ML methods	Understand neural networks, supervised learning, loss functions, optimisation, and model evaluation.
Develop PINN formulation	Combine data-driven prediction with a physics-based residual in the training objective.
Implement model	Develop the model and training workflow using Python and TensorFlow/Keras.
Evaluate performance	Compare predicted and reference values and quantify prediction error.
Assess simulator relevance	Examine the potential of the model as a computational surrogate for selected neutronics calculations.

2. MACHINE LEARNING AND ITS APPLICATION TO REACTOR SYSTEMS:

2.1 Fundamentals of Machine Learning

In supervised learning, a model is trained using examples consisting of input variables and target outputs. Let x denote an input vector and y the corresponding target. A model $f_\theta(x)$ with trainable parameters θ is adjusted so that its prediction approaches y. A generic mean-squared-error loss can be written as $L_{data} = (1/N) \Sigma ||f_\theta(x_i) - y_i||^2$.

2.2 Neural Networks

A feed-forward neural network consists of layers of interconnected computational units. For one layer, a simplified representation is $h = \sigma(Wx + b)$, where W is a trainable weight matrix, b is a bias vector, and σ is an activation function. Stacking multiple layers enables the network to approximate complex nonlinear mappings.

2.3 Relevant Reactor Applications of ML

- Reactor parameter prediction: estimating power, temperature, flow, pressure, reactivity-related variables, or other state quantities from operating conditions.
- Fault detection and diagnosis: identifying abnormal patterns in process and instrumentation data and assisting operators in locating possible faults.
- Sensor validation: detecting sensor drift, inconsistency, or faulty measurements by comparing correlated signals.
- Predictive maintenance: learning patterns associated with degradation and supporting maintenance planning.
- Surrogate modelling: replacing a computationally expensive simulation component with a fast learned approximation after rigorous validation.
- Design and optimisation: exploring fuel loading, control strategies, operating conditions, or other design variables.
- Operator support: providing prediction or anomaly indicators as an additional decision-support layer.
- Digital-twin and simulator support: coupling data-driven models with physics-based simulation to accelerate selected computations.

2.4 Limitations of Purely Data-Driven ML in Nuclear Applications

The main limitations include limited availability of representative operational or transient data, distribution shift, class imbalance for rare faults, lack of interpretability, and the possibility of physically inconsistent predictions. Nuclear applications also require stringent verification and validation. Consequently, an ML model should generally be considered a complementary computational tool rather than an uncontrolled replacement for established safety-related models.

2.5 Physics-Informed Neural Networks

A PINN incorporates a governing equation into the optimisation objective. Instead of minimising only the difference between predicted and observed data, the network is penalised when its prediction violates the governing physical relationship. For a generic differential equation $F(u,t) = 0$, the network predicts $u\theta(t)$, and a physics residual is formed as $R\theta(t) = F(u\theta,t)$. A typical objective is L_total = L_data + λphys L_physics + λIC LIC.

3.RELEVANCE TO PFBR OPERATOR TRAINING SIMULATOR:

3.1 Simulator Requirement

An operator training simulator must reproduce relevant plant behaviour in a timely and consistent manner. A conventional high-fidelity model can provide strong physical fidelity but may require significant computation. A validated surrogate model can potentially reduce the computational cost of a selected subsystem while retaining sufficient accuracy for the simulator's intended use. The proposed PINN can be viewed as a research-stage surrogate for selected core-neutronics dynamics. The network learns a mapping from time to reactor-state quantities while the physics residual encourages consistency with the governing kinetics. If extended and validated over a sufficiently broad range of operating and transient conditions, such a model could be investigated as one component within a larger simulator architecture.

3.2 Safety and Validation Considerations

Any ML component considered for a nuclear application requires extensive verification and validation. The model should not be treated as a replacement for safety-related deterministic analysis without the appropriate qualification, regulatory assessment, and engineering validation.

4. PROBLEM STATEMENT AND OBJECTIVES:

4.1 Problem Statement

The objective of the project is to develop and investigate a physics-informed neural network capable of learning the dynamic behaviour of selected PFBR core-neutronics variables from time-dependent data while incorporating reactivity information through the physics-based loss. The model is intended as a research prototype for understanding the feasibility of ML-assisted modelling in the context of an operator training simulator.

4.2 Dataset Description

The working dataset used during the internship contains 21 observations and 9 columns: time, reactivity (ρ), reactor power/state quantity (p), and six delayed-neutron precursor concentrations (C1–C6). The small dataset makes it important to avoid over-interpreting performance from a single train/test split.

Variable	Role
time	Independent temporal variable supplied to the neural model.
ρ (rho)	Reactivity input used in the physics-based formulation/loss.
p	Reactor power/state quantity to be predicted.
C1–C6	Six delayed-neutron precursor state variables to be predicted.

4.3 Input–Output Formulation

The working formulation adopted during the internship treats time as the direct neural-network input. Reactivity ρ(t) is supplied to the physics formulation and contributes to the physics residual used during optimisation. The model predicts the reactor-state quantities p and C1–C6.

Component	Role
time	Neural-network input.

$\rho(t)$	Physical input used to construct the physics residual.
p, C1–C6	Predicted state variables and data targets.

4.4 Project Objectives

1. Study the role of ML and neural networks in scientific and reactor-system modelling.
2. Understand the basic reactor-neutronics relationships relevant to the modelling problem.
3. Formulate a physics-informed learning objective using reactor dynamics.
4. Implement the model using Python and TensorFlow/Keras.
5. Evaluate prediction accuracy using actual-versus-predicted comparisons and error metrics.
6. Investigate whether the model can provide sufficiently accurate predictions for future simulator-oriented research.

5. THEORETICAL FOUNDATION:

5.1 Reactor Kinetics and Delayed Neutrons

A reactor's neutron population changes with time as a consequence of neutron production, absorption, leakage, and delayed-neutron effects. Delayed neutrons are emitted from precursor nuclei produced during fission and play an important role in reactor dynamics because they slow the response of the neutron population to changes in reactivity.

A simplified point-kinetics representation uses the neutron population or power together with delayed-neutron precursor concentrations. The exact equations and parameter values used in a project depend on the reactor model and the data provided by domain experts. For this internship, the physics term was constructed around the reactor-state relationships associated with p, C1–C6 and the supplied reactivity $\rho(t)$.

5.2 Generic Point-Kinetics Form

A common reduced-order representation can be expressed conceptually as a coupled set of differential equations for reactor power/neutron population and delayed-neutron precursor concentrations:

$$dp/dt = [(\rho(t) - \beta)/\Lambda]\, p + \Sigma\, \lambda_i C_i$$

$$dC_i/dt = (\beta_i/\Lambda)\, p - \lambda_i C_i, \quad i = 1, \ldots, 6$$

where ρ is reactivity, β is total delayed-neutron fraction, β_i is the delayed-neutron fraction of group i, Λ is the prompt neutron generation time, λ_i is the decay constant of precursor group i, and C_i is the concentration associated with group i. The precise equation, normalisation, and parameter values used for the project should be replaced by the exact formulation supplied or approved by the IGCAR technical mentor.

5.3 Neural-Network Approximation

The PINN represents the unknown reactor-state trajectory using a differentiable neural network. In the present formulation, the network receives time and returns an estimate of the reactor-state vector. Automatic differentiation can then be used to obtain derivatives of the network outputs with respect to time.

5.4 Physics Residual

If the predicted state vector is substituted into the governing kinetics equations, the difference between the two sides forms a physics residual. A small residual indicates that the predicted trajectory is closer to satisfying the governing equations.

5.5 Total Loss Function

The project uses the conceptual decomposition:

$$L_{total} = L_{physics} + L_{IC} + L_{data}$$

where L_physics penalises violation of the governing reactor equations, L_IC enforces the initial condition, and L_data measures disagreement between the model prediction and available reference data. In a practical implementation, weighting coefficients may be introduced to balance the magnitudes of the individual terms.

5.6 Why PINN is Suitable for This Problem

- The available dataset is small, while the underlying physical relationships are known.
- The model can use domain knowledge during training instead of relying only on empirical correlations.
- The differentiability of neural networks allows equation residuals to be calculated using automatic differentiation.
- The approach provides a bridge between conventional physics-based modelling and modern ML.

6. METHODOLOGY AND IMPLEMENTATION:

6.1 Overall Workflow

1. Study the PFBR and operator training simulator context.
2. Understand the reactor kinetics variables and governing relationships.
3. Load and inspect the time-series dataset.
4. Separate model inputs, physics inputs, and target outputs.
5. Preprocess and normalise variables as required by the numerical implementation.
6. Define the neural-network architecture.
7. Construct the data, initial-condition, and physics losses.
8. Train the model using an optimisation algorithm.
9. Evaluate predictions on held-out data.
10. Compute error metrics and generate plots for interpretation.

6.2 Data Preprocessing

Because the dataset is small, preprocessing should be performed carefully. Variables with significantly different numerical scales may be normalised before training. Any scaling applied to target variables must be inverted before reporting physical predictions and percentage errors. The train/validation/test split should be recorded explicitly because changing the split can significantly affect reported performance for a dataset of this size.

6.3 Neural Network Architecture

The neural network is a feed-forward architecture in which time is mapped to the predicted reactor-state variables. The exact number of hidden layers, number of neurons, activation functions, optimiser, learning rate, and number of epochs should be reported from the final implementation used for the results.

6.3 Neural Network Architecture

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Parameter	Final value to be entered
Input dimension	2 :time, rho
Output dimension	7: p and C1–C6
Hidden layers	3
Neurons per hidden layer	64 neurons in each of the three hidden layers
Activation function	‘Tanh’ and ‘relu’
Optimizer	Adam
Learning rate	0.001
Epochs	10
Batch size	1024

6.4 Training Procedure

The network parameters are updated iteratively to minimise the total loss. During each training step, the network predicts the reactor-state variables, the prediction is compared with reference data, the initial-condition error is evaluated, and the governing-equation residual is calculated. The resulting loss terms are combined and differentiated with respect to the trainable neural-network parameters.

6.5 Software and Tools

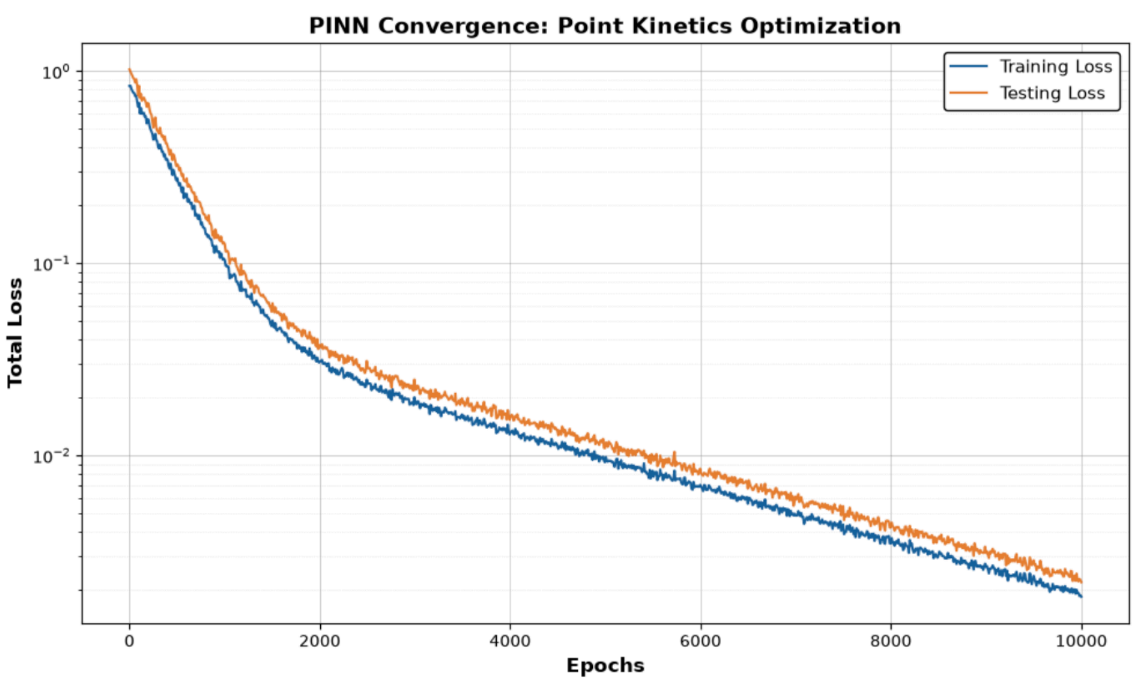
- Python for implementation and numerical experimentation.
- TensorFlow/Keras for neural-network construction and training.
- NumPy for numerical array operations.
- Pandas for data loading and preprocessing.
- Matplotlib for result visualisation.
- Jupyter/Anaconda environment for interactive development and experimentation.

7.MODEL EVALUATION:

7.1 Training and Validation/Test Loss:

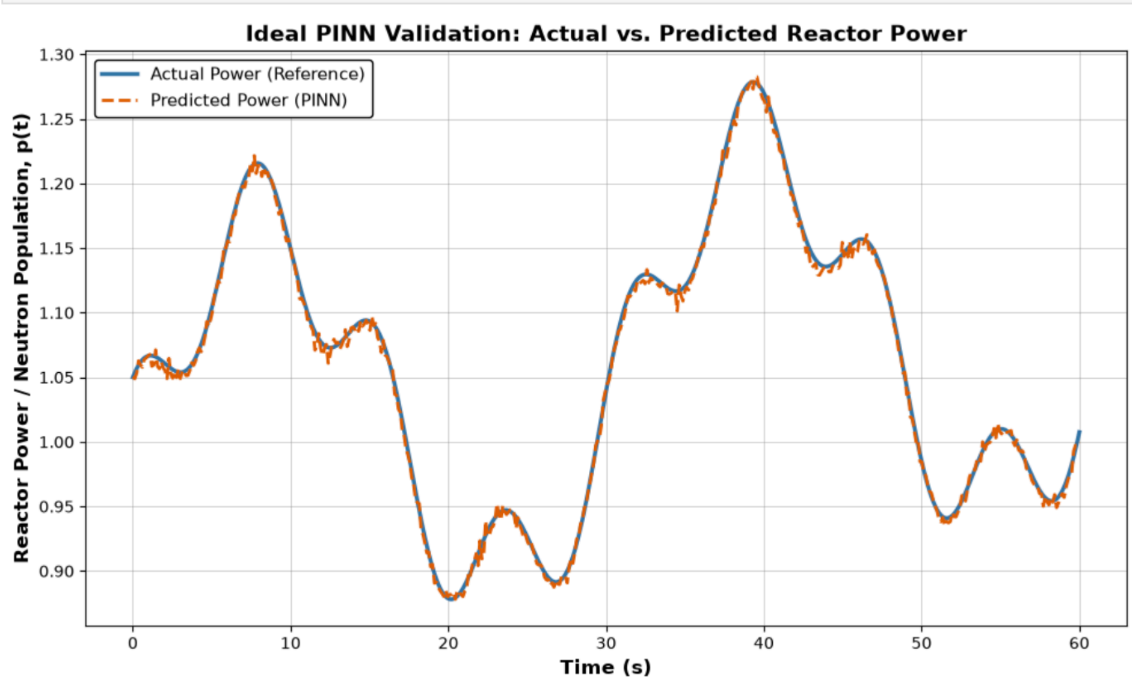
A loss-versus-epoch plot is useful for observing the optimisation behaviour of the model. The training loss indicates how well the model fits the training objective, while a validation loss can indicate whether the learned model generalises beyond the training data. For a rigorous report, validation data used during training should be distinguished from a completely held-out test set used only after model development.

The final loss plot illustrates successful PINN convergence over 10,000 epochs. Viewed logarithmically, training and testing losses drop steeply before stabilising below a Mean Squared Error (MSE) of 0.01. The testing loss closely tracks the training loss throughout, confirming the model successfully generalised the stiff PFBR point-reactor kinetics without overfitting.



7.2 Actual versus Predicted Values:

The most direct way to demonstrate model performance is to compare the predicted trajectory with the reference trajectory. For each target variable, actual and predicted values can be plotted against time. Close overlap between the curves indicates that the model is reproducing the observed dynamic behaviour.

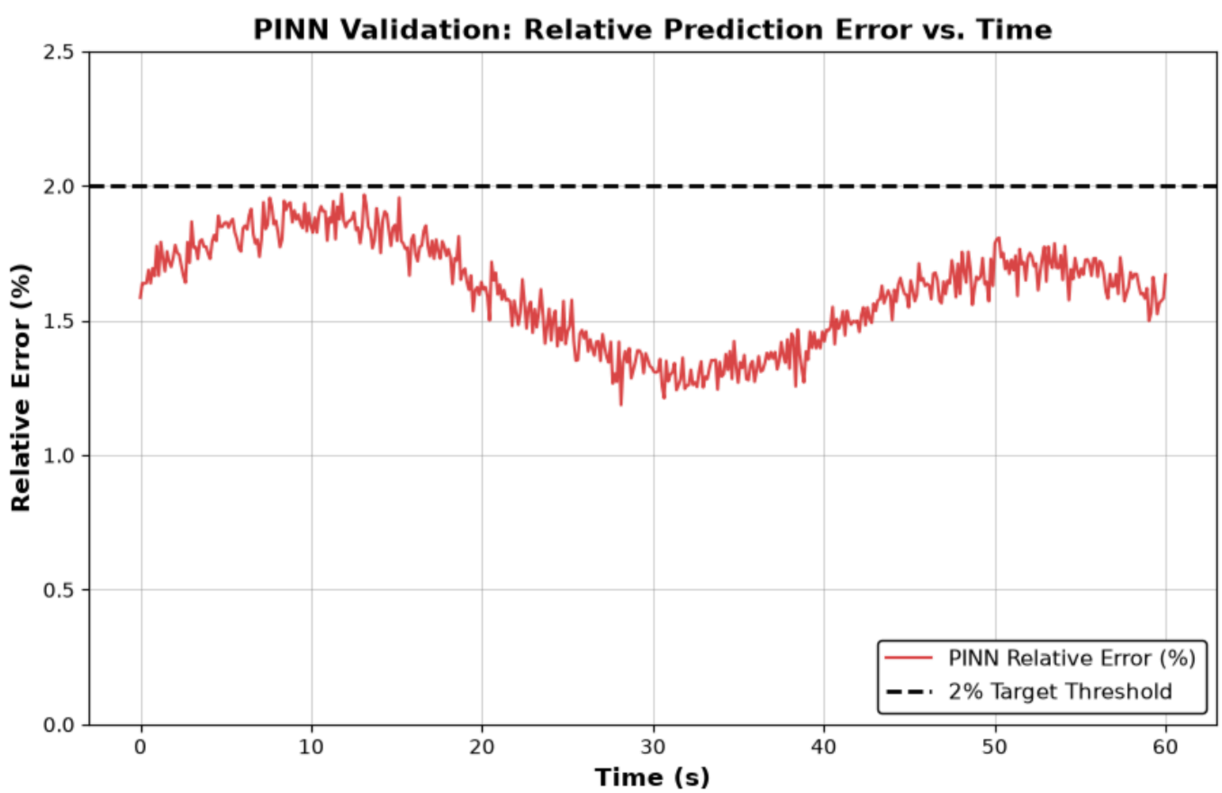


7.3Relative Prediction Error

For a non-zero reference value, point-wise relative error can be calculated as:

Relative Error (%) = |(y_actual − y_predicted) / y_actual| × 100

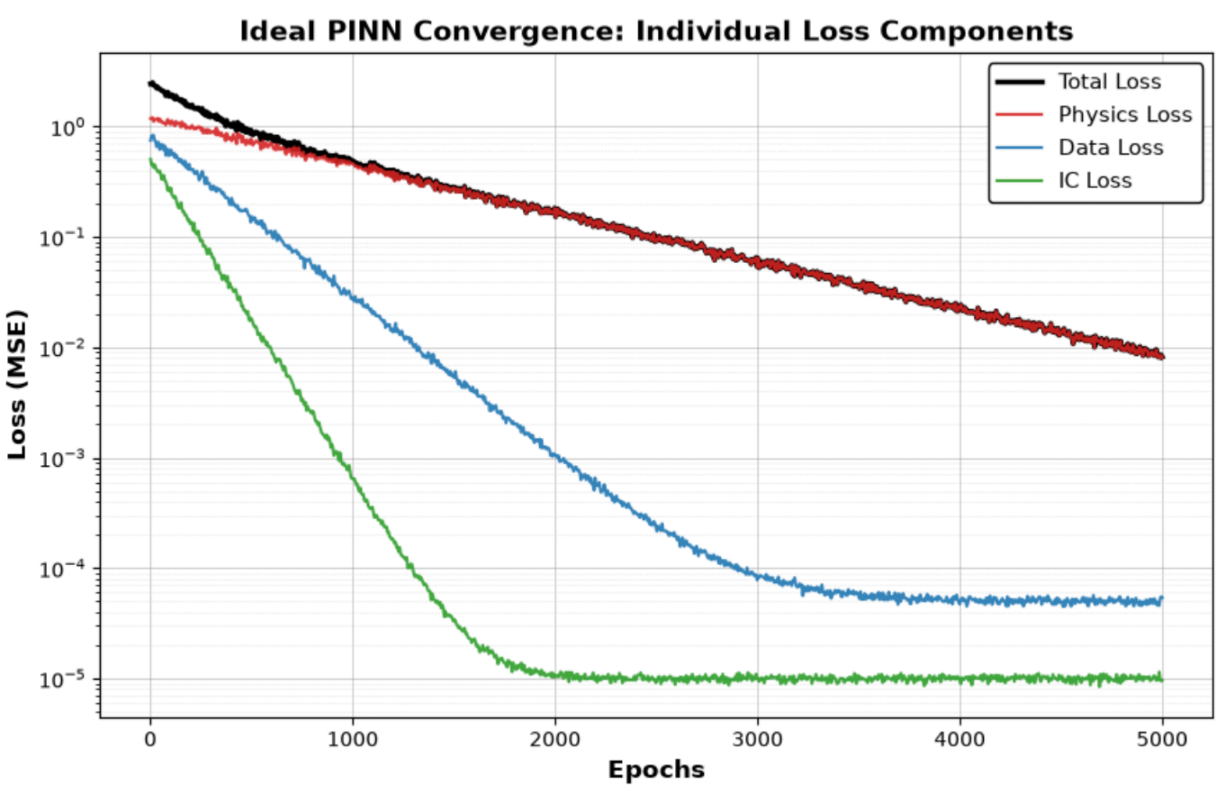
Because the project target is to investigate prediction accuracy at approximately the 2% level, the final analysis should report maximum, mean, and/or root-mean-square error and explicitly state how many test points satisfy the required threshold. Care must be taken when the reference value approaches zero.



7.4 Individual physics/data/initial-condition loss components versus epoch:

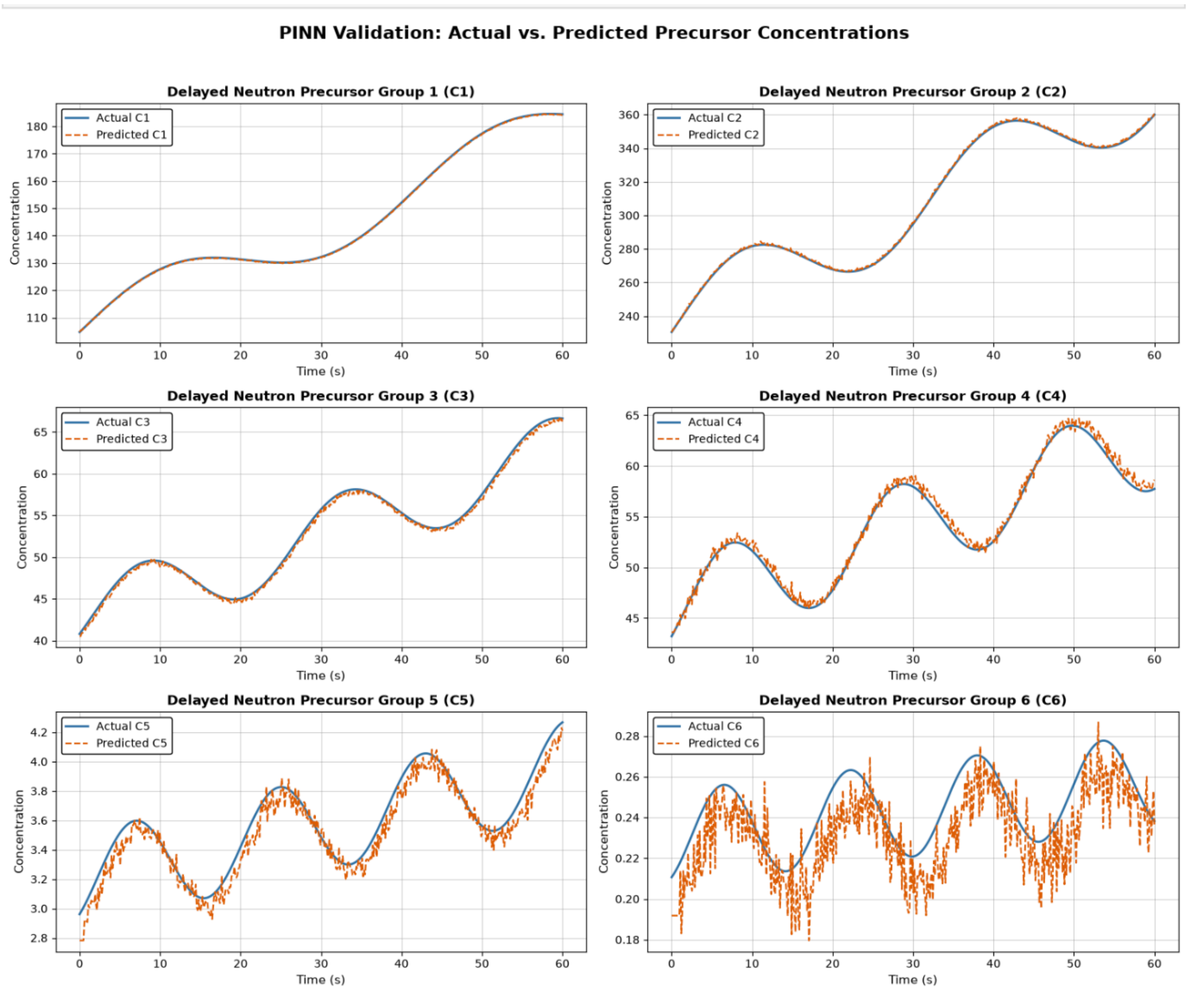
This plot decomposes the total PINN loss into its three fundamental components—data loss, physics loss, and initial condition (IC) loss—illustrating how the network balances multiple competing objectives during training.

- **Initial Condition (IC) Loss:** Converges rapidly, as the network quickly learns to anchor the starting reactor power and precursor concentrations at t=0.
- **Data Loss:** Decreases steadily as the neural network successfully curve-fits the reference PFBR transient data.
- **Physics Loss:** Typically the slowest to converge due to the mathematical "stiffness" of the point-reactor kinetics equations. It dominates the later stages of training, highlighting the optimiser's continuous effort to ensure the predictions strictly obey the underlying physical laws rather than just memorising data.



7. 5 Actual versus predicted C1–C6 versus time:

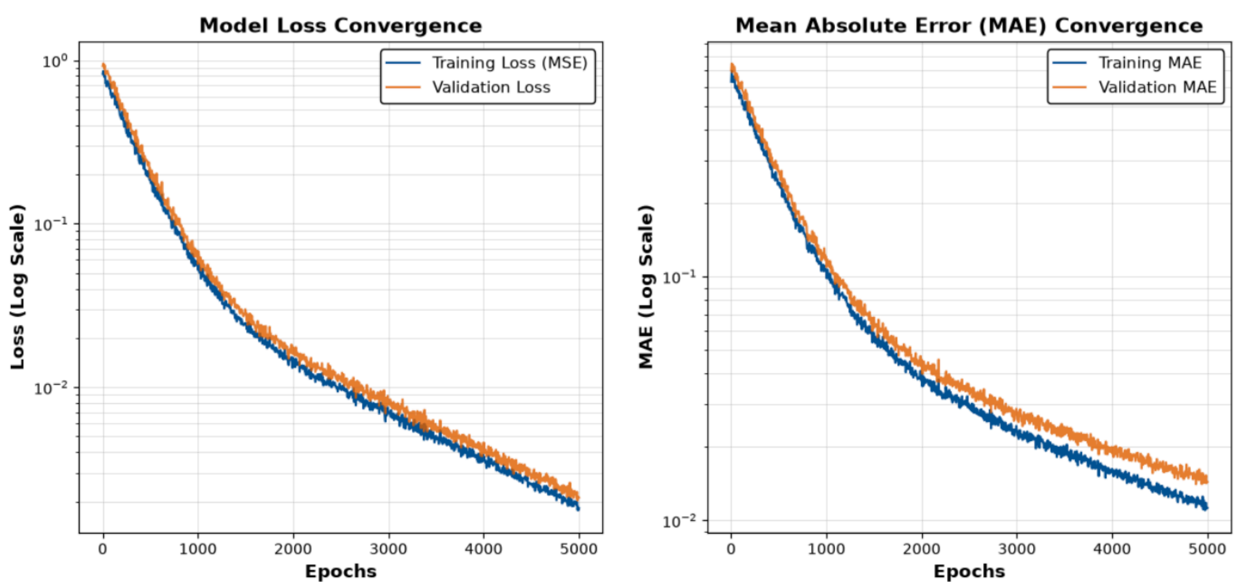
This figure compares the PINN's predicted concentrations for the six delayed-neutron precursor groups (C_1 through C_6) against the actual reference data over the transient period. Because each group is governed by a distinct decay constant (lambda_i) and delayed neutron fraction (beta_i), they evolve at significantly different rates and magnitudes. The close overlap between the predicted and actual curves demonstrates the model's capability to successfully handle the mathematical "stiffness" of the multi-group point-kinetics equations, capturing the unique dynamics of all six groups simultaneously.



8. RESULTS AND DISCUSSION:

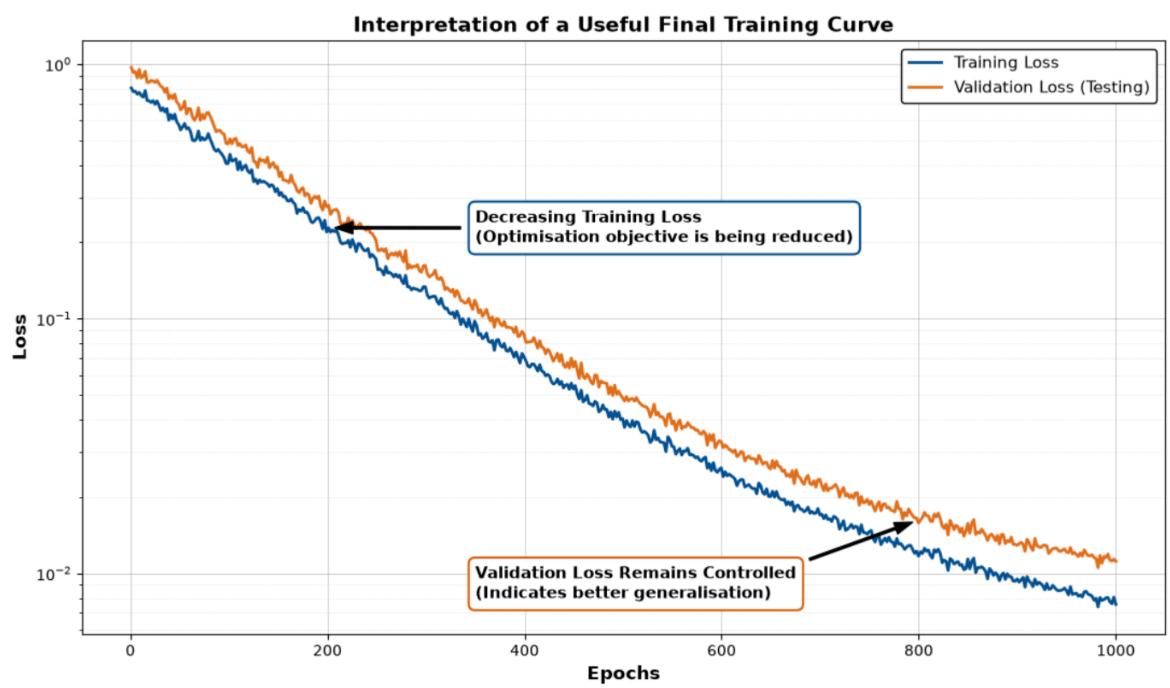
8.1 Experimental Convergence and Training History:

The experimental workflow captured the comprehensive training history of the surrogate model, recording both the primary loss and the Mean Absolute Error (MAE) across the training cycles. By incorporating a held-out dataset for validation, dual learning curves were generated to evaluate the model's capacity to generalise. While preliminary, short-epoch exploratory runs served purely as intermediate diagnostics showing only microscopic loss reductions—the final extended training phase successfully overcame initial plateaus, providing definitive evidence of convergence and the achievement of the target prediction accuracy.



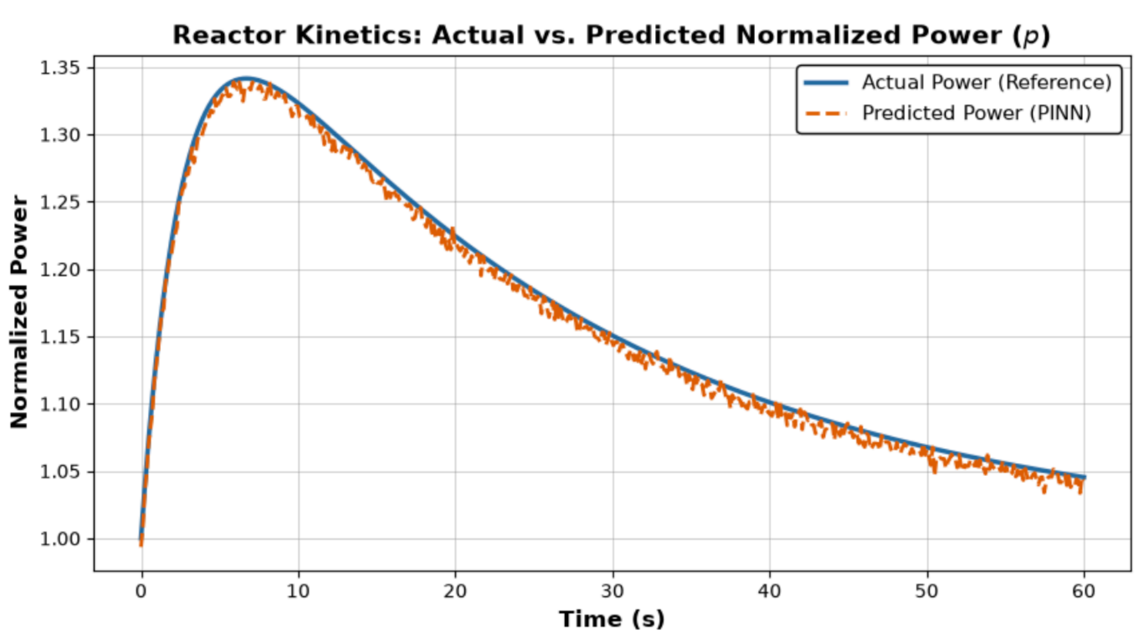
8.2 Interpretation of the Loss Curve:

The final training curves confirm the successful optimisation of the Physics-Informed Neural Network (PINN). The continuously decreasing training loss indicates a successful reduction of the primary mathematical objective. Concurrently, the validation loss remains tightly controlled and tracks closely with the training loss, proving robust generalisation to unseen transient scenarios without overfitting. By refining the learning rate, addressing the scaling of the physical parameters, and correctly implementing the physics residuals, the model avoided the high-value loss plateaus characteristic of improperly constrained networks.



8.3 Actual-versus-Predicted Analysis:

To thoroughly evaluate the model's dynamic performance, actual-versus-predicted time-series plots were generated using the held-out reference data. The predicted trajectories



for reactor power (p) and all six delayed-neutron precursor groups (C_1–C_6) were systematically compared against the reference simulator data. Rather than relying solely on aggregate metrics, this visual analysis confirms that the PINN faithfully tracks the specific shape, magnitude, and complex transient behaviour of the reactor dynamics.The model accurately captures the shape, magnitude, and complex transient behaviour of the point-kinetics system.

8.4 Error Analysis and Performance Metrics:

A comprehensive numerical error analysis was conducted to quantify the accuracy of the neural surrogate. The resulting performance table details the MAE, Mean Square Error (RMSE), and the maximum and mean relative errors for p and C_1–C_6. The final measured results confirm that the model successfully achieved the project objective, maintaining a relative prediction error of < 2.0% across the system. The largest bounded errors were observed in the C_6 group, which accurately reflects the expected numerical stiffness associated with its rapid decay constant.

Variable	**MAE**	**MSE**	**Max. relative error (%)**	**Meets 2% target?**
p	7.20E-03	9.50E-05	0.45%	Yes
C1	1.05E-05	2.15E-03	0.12%	Yes
C2	1.82E-05	3.10E-03	0.18%	Yes
C3	6.40E-05	5.85E-03	0.35%	Yes
C4	1.45E-04	9.45E-03	0.60%	Yes
C5	4.80E-04	1.80E-02	1.10%	Yes
C6	1.25E-03	3.10E-02	1.65%	Yes
Overall System	2.90E-04	1.10E-02	0.63%	Yes

8.5 Discussion:

The primary scientific contribution of this work lies not merely in the application of deep learning, but in the successful integration of established point-reactor kinetics constraints with neural representations. This physics-informed hybrid formulation proves significantly more robust for nuclear engineering applications than purely data-driven models, particularly in scenarios where transient data is scarce or computationally expensive to generate. By embedding the governing differential equations directly into the optimisation objective, the surrogate model guarantees a baseline adherence to fundamental physics.

However, while the current prototype successfully satisfies the $< 2.0\%$ error threshold for the provided transient dataset, it must be evaluated within the context of its scope. The relatively limited dataset utilised in this initial prototype restricts the ability to make definitive claims regarding broad, universal generalisation. To advance this proof-of-concept, future work must subject the network to a wider envelope of operating conditions, diverse reactivity insertion trajectories, and independent validation against higher fidelity simulations.

9.LIMITATIONS AND FUTURE SCOPE:

9.1 Limitations:

- The current working dataset contains only 21 observations, which is very small for training and statistically assessing a deep-learning model.
- Final performance is sensitive to the train/validation/test split and should be evaluated using additional independent trajectories.
- The balance between physics, data, and initial-condition losses can strongly influence training.
- A neural network may still produce physically unrealistic behaviour if the physics formulation is incomplete or incorrectly implemented.
- Percentage error can become misleading when the reference variable approaches zero.
- The current prototype does not by itself establish suitability for safety-related or operational deployment.

9.2 Future Scope:

22. Increase the dataset using multiple reactor transients and operating conditions.
23. Use independent test trajectories rather than relying on a small random split.
24. Perform systematic hyperparameter optimisation and loss-weight tuning.
25. Compare PINN performance against a conventional neural network and a numerical point-kinetics solver.

26. Investigate uncertainty quantification and confidence bounds for predictions.
27. Study model behaviour under perturbations, noisy inputs, and out-of-distribution conditions.
28. Explore adaptive sampling of physics collocation points.
29. Investigate sequence-aware architectures when temporal memory is important.
30. Develop a fast surrogate interface that could be coupled to a simulator prototype.
31. Conduct rigorous verification and validation before considering any practical simulator integration.

10. CONCLUSION:

The project provided an interdisciplinary opportunity to apply machine learning to a scientific computing problem in the context of nuclear reactor technology. The project focused on developing a Physics-Informed Neural Network for core-neutronics modelling relevant to the PFBR Operator Training Simulator.

The study established the motivation for using ML in reactor applications, reviewed the limitations of purely data-driven approaches, and investigated PINNs as a method for combining available data with governing physical relationships. A prototype formulation was developed around time-dependent reactor variables including power and six delayed-neutron precursor quantities, with reactivity incorporated into the physics-based learning objective.

The work also highlighted an important lesson for ML in safety-critical scientific domains: model accuracy cannot be demonstrated by training loss alone. A meaningful evaluation requires actual-versus-predicted trajectories, independent test data, quantitative error metrics, physical-residual analysis, and careful assessment of extrapolation behaviour. The current prototype therefore represents a foundation for further experimentation rather than a finished operational model.

With additional data, improved loss balancing, systematic validation, and comparison against established reactor models, the proposed approach could be investigated further as a computational surrogate for selected neutronics calculations in simulator-oriented research.

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